

Comparative Analysis of Multi-Face Detection Methods in Classroom Environments: Haar Cascade, MTCNN, YOLOFace, and RetinaFace

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Abstract— Face detection in educational environments is vital for applications such as automated attendance systems, behavior analysis, and personalized learning tools. This study conducted a comprehensive comparative analysis of four advanced face detection algorithms—Haar Cascade, MTCNN, YOLOFace, and RetinaFace—using a dataset of real classroom images. The analysis addresses the unique challenges of classroom settings, including variable lighting, occlusions, and diverse student orientations. Metrics such as precision, recall, F1-score, and inference time were evaluated. YOLOFace and RetinaFace demonstrated superior performance, with YOLOFace achieving perfect precision, recall, and F1-score in most scenarios. RetinaFace also demonstrated high precision and recall, though its inference times were longer, up to 12.73 seconds. MTCNN exhibited high precision, up to 1.00, and recall, up to 0.97, but had higher inference times and occasionally missed some faces. Haar Cascade, while efficient with the shortest inference times, down to 0.54 seconds, displayed lower precision of 0.89 and recall of 0.25, resulting in a higher rate of missed detections. These findings emphasize the importance of selecting appropriate face detection models that address the specific challenges of educational environments to enhance the effectiveness of digital classroom applications.

Keywords—Comparative Analysis, Face detection, classroom environment

I. INTRODUCTION

In the field of educational technology, face detection plays a pivotal role in enhancing classroom experiences through applications such as automated attendance systems, student engagement tracking, and personalized learning interventions [1]. By automating attendance, schools can significantly reduce the administrative workload for educators, offering a more efficient alternative to replace traditional methods like paper-based registers, fingerprint scanning, or single-entry face recognition systems. Furthermore, facial recognition can provide real-time insights into student engagement, enabling educators to identify students who may need additional support or are disengaged, thus fostering a more inclusive and effective learning environment [2], [3], [4]. Consequently, implementing a robust and reliable face detection system is essential to ensuring the accuracy and effectiveness of these technologies [5].

Previous studies have investigated student monitoring in classrooms, revealing that recognition accuracy remains unsatisfactory [6]. In response to these challenges, various efforts have been made to improve face detection methods, including optimizing YOLOv3 using Bayesian techniques and developing the MTCNN algorithm [5], [7]. These coordinated efforts highlight the need for further research on face detection in classrooms. Therefore, a comparative approach that examines different face detection methods is important for finding the best ways to improve recognition accuracy in educational settings.

Classroom environments present unique challenges for face detection, such as fluctuating lighting, frequent occlusions, and varying student orientations. These dynamic conditions necessitate the development and evaluation of advanced detection algorithms tailored for educational settings [7]. This study compares four face detection models: Haar Cascade, MTCNN, YOLOFace, and RetinaFace, tested in real classroom situations. Haar Cascade, known for being fast and lightweight, struggles with high false-negative rates in difficult conditions. On the other hand, modern models like MTCNN, YOLOFace, and RetinaFace use deep learning to achieve better accuracy and reliability.

Although many face detection models were validated on widely-used datasets, such as Labeled Faces in the Wild (LFW), FDDB, and WIDER FACE [8], their performance in classroom settings, with its inherent complexities, remains underexplored [9]. This study bridges this gap by testing these models in real classrooms, which are characterized by fluctuating lighting and diverse student postures that are not often present in commonly used benchmark datasets. By conducting a comparative analysis of both traditional and deep learning-based approaches, this study offers a comprehensive evaluation of their suitability for real-time use in educational environments. The insights gained from this research are expected to inform the development of more accurate, efficient, and reliable automated attendance systems and student engagement tracking tools, ultimately contributing to the advancement of face detection technologies specifically tailored for educational purposes.

II. RELATED WORK

Face detection has advanced significantly over the years, with progress shifting from traditional methods to more sophisticated deep learning-based techniques. Figure 1 summarizes the categorization of face detection methods discussed in this study, as well as some related research that addresses the application of face detection in educational environments.

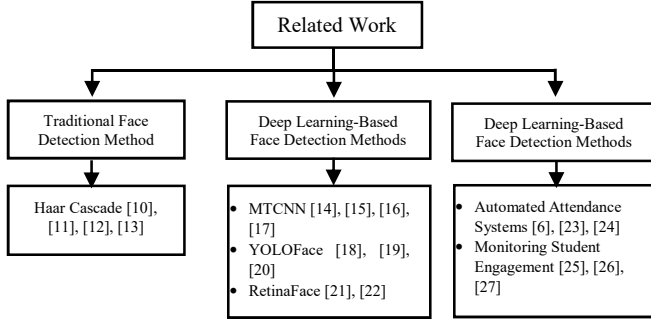


Fig 1. Related Work Diagram

A. Traditional Face Detection Methods

The Haar Cascade classifier, developed by Viola and Jones [10], is one of the earliest and most influential face detection algorithms. This algorithm uses a series of simple features, evaluated using integral images, to detect faces with real-time performance. Numerous previous studies have utilized and evaluated this algorithm, including research that compared the performance of Haar Cascade with other face detection methods and studies that applied Haar Cascade under various lighting conditions [11], [12]. Additionally, research has examined the effectiveness of Haar Cascade in various face recognition applications, such as head pose, facial expressions, image orientation, occlusion, and lighting [13]. These studies indicate that, despite some limitations—such as less accurate detection of non-frontal faces—Haar Cascade remains a popular choice for basic face detection and real-time applications.

B. Deep Learning-Based Face Detection Methods

With advancements in deep learning, various sophisticated face detection models have emerged, significantly enhancing the accuracy and reliability of face detection. One such method introduced by Zhang et al. is Multi-task Cascaded Convolutional Networks (MTCNN) [14]. This method excels in detecting faces and facial landmarks with high precision through several stages of convolutional neural networks [15]. MTCNN has proven effective in handling variations in scale, pose, and facial expressions, making it a robust choice for diverse environments [16], [17].

In contrast, YOLOv3 (You Only Look Once, version 3) offers a real-time object detection approach by applying a single neural network to the entire image, dividing it into regions, and predicting bounding boxes and probabilities for each region simultaneously [18]. This approach has shown impressive results in terms of speed and accuracy for face detection, though it requires substantial computational resources [19].

YOLOFace is an adaptation of YOLOv3 specifically designed for face detection tasks [20]. This variant modifies the YOLOv3 architecture to enhance its ability to detect faces

more accurately under various conditions [20]. Although there is no formal academic paper available for YOLOFace, the GitHub repository provides detailed implementation and usage guidelines.

Finally, RetinaFace introduces innovation by integrating pixel-level face localization tasks into the detection process, significantly improving detection performance [21]. Utilizing a deep residual network backbone, RetinaFace achieves state-of-the-art accuracy on several benchmark datasets such as WIDER FACE and FDDB, demonstrating its effectiveness in handling challenging detection conditions [22].

C. Face Detection in Educational Settings

Several studies have explored the use of face detection in educational contexts, particularly focusing on automated attendance systems and student engagement analysis within classrooms. For instance, one study implemented an automated attendance system using facial recognition, which significantly reduced administrative tasks for educators [6], [23], [24]. Similarly, another study developed a system to track student engagement in real-time, providing valuable insights for personalized learning interventions [25], [26], [27].

Despite these advancements, there remains a gap in the comprehensive evaluation of face detection algorithms specifically within classroom environments. The unique challenges of these settings, including varying lighting conditions, occlusions, and the presence of many students, necessitate further research to identify the most effective face detection models for practical applications.

III. METHOD

This section outlines the methodology used to compare four face detection models: Haar Cascade, MTCNN, YOLOFace, and RetinaFace. Images were collected from real classroom settings with varying conditions to simulate practical environments. The models were tested for performance, focusing on detection speed and accuracy. Intersection over Union (IoU) was used to evaluate the precision of face localization. Key metrics, including Precision, Recall, F1-score, and Inference time, were calculated to assess overall performance. The flowchart methodology can be seen in Figure 2.

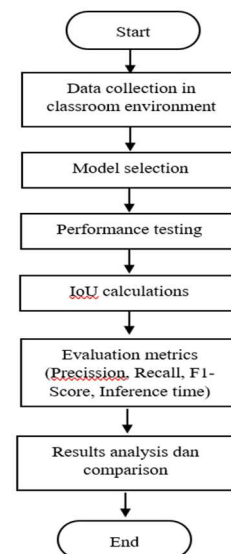


Fig 2. Flowchat methodology

After completing the evaluation, the results were analyzed in terms of performance across various conditions, including different levels of occlusion, lighting variations, and the number of faces in the classroom. The analysis focused on both the detection accuracy and the computational efficiency of each model, allowing for an informed comparison that could guide future implementations in educational settings.

A. Dataset

In this study, a specialized dataset was collected from various classroom environments, capturing diverse conditions such as different lighting, occlusions, and student orientations. The four models used in this research were tested on four test datasets, each representing a different classroom scenario. A summary of the four test scenarios used is shown in Table 1 and Figure 3.

TABLE I. THE FOUR DATATEST SCENARIO

Scenario	Number of student	Illuminance (Lux)
1	40	152
2	38	152
3	32	152
4	32	40



Fig 3. Images used in the dataset for each scenario

B. Model Selection

Pre-trained models were employed directly for face detection on the collected dataset, without additional training or fine-tuning, to evaluate their performance in their original states. The models selected for this study include Haar Cascade, MTCNN, YOLOFace, and RetinaFace.

Haar Cascade, sourced from OpenCV, is utilized for its lightweight computational requirements, which makes it suitable for real-time applications. However, it may face limitations under challenging conditions such as varied lighting and angles. MTCNN, using a pre-trained model from the mtcnn library, is chosen for its superior performance in detecting faces and facial landmarks with high precision, particularly in diverse and complex scenarios. YOLOFace, obtained from the Yoloface repository, is recognized for its speed and accuracy in object detection, including face detection, making it ideal for applications that require rapid processing. Lastly, RetinaFace, implemented with a pre-trained model from the retina-face library, is selected for its advanced pixel-level face localization capabilities, which significantly enhance detection performance and robustness across various face sizes and conditions.

C. Evaluation

The performance of each face detection model was evaluated using several key metrics, calculated through Intersection over Union (IoU) and ground truth data. IoU measures the overlap between the predicted bounding boxes (B_p) and the ground truth bounding boxes (B_{gt}) [28]. It was calculated as the ratio of the area of intersection to the area of union of the two bounding boxes, indicating how well the model predicts the location and size of faces. The calculation of IoU are shown in Equation 1.

$$IoU = \frac{area(B_p \cap B_{gt})}{area(B_p \cup B_{gt})} \quad (1)$$

This can be illustrated as shown in Figure 4.

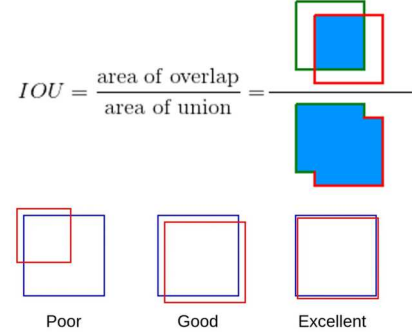


Fig 4. The illustration and calculation of IoU [28]

By comparing the IoU with a given threshold (t), we can classify a detection as either correct or incorrect. If the IoU is greater than or equal to the threshold $IoU \geq t$, the detection is considered correct (True Positive). If the IoU is less than the threshold $IoU < t$, the detection is considered incorrect (False Positive) [28]. Additionally, a False Negative (FN) occurs when an object is present in the ground truth but the model fails to detect it or the detection does not meet the IoU threshold, leading to a missed detection. Commonly, a threshold between 0.5 and 0.75 is used [29]. Thus, Precision, Recall, and F1-score can be calculated using the following equations.

Using information about True Positives (TP), False Positives (FP), and False Negatives (FN) obtained from IoU calculations, evaluation metrics such as Precision, Recall, and F1-Score were computed. Precision measures the proportion of true predictions among all predictions made by the model, Recall measures the proportion of ground truth objects successfully detected by the model, and F1-Score was the harmonic mean of Precision and Recall, providing a balanced view of the model's performance. The evaluation also includes measurements of processing speed, robustness under various conditions, and face localization quality to ensure the model performs well in diverse situations.

IV. RESULTS AND DISCUSSIONS

The comparative analysis of four face detection algorithms—Haar Cascade, MTCNN, YOLOFace, and RetinaFace—was conducted across four distinct test scenarios. Experiments were performed on Google Colab with a runtime equipped with an NVIDIA T4 GPU. The performance metrics, including Precision, Recall, F1-Score, and Inference Time, were evaluated to determine the efficacy

and efficiency of each algorithm in detecting faces within classroom environments.

The visual results of the face detection algorithms in the classroom environment for one example test scenario, specifically test scenario 1, are shown in Figure 5. Each image represents the output of the face detection models for this specific test scenario.



Number of faces detected: 14

(a)



Number of faces detected: 36

(b)



Number of faces detected: 40

(c)



Number of faces detected: 40

(d)

Fig 5. Test results in scenario 1 (a) Haar cascade (b) MTCNN (c) YOLOface (d) Retinaface

The results, as visualized in the images provided, show the bounding boxes predicted by each model (in green) and the ground truth annotations (in blue). The detailed metrics for precision, recall, F1-score, and inference time are summarized in the table 2.

TABLE II. EVALUATION METRIC ON FOUR SCENARIOS

Test Scenario	Method	Precision	Recall	F1-score	Inference Time (s)
1	Haar Cascade	0.93	0.33	0.48	0.5409
	MTCNN	1	0.9	0.95	4.6908
	YOLOFace	1	1	1	2.7569
	RetinaFace	1	1	1	9.146
2	Haar Cascade	0.93	0.34	0.5	0.7535
	MTCNN	1	0.97	0.99	10.084
	YOLOFace	1	1	1	2.1548
	RetinaFace	0.97	1	0.99	12.7296
3	Haar Cascade	0.92	0.34	0.5	0.803
	MTCNN	0.97	0.97	0.97	4.271
	YOLOFace	1	1	1	2.259
	RetinaFace	1	1	1	9.6616
4	Haar Cascade	0.89	0.25	0.39	0.5511
	MTCNN	1	0.84	0.92	5.5228
	YOLOFace	1	0.97	0.98	3.3302
	RetinaFace	1	1	1	11.4697

In our comparative analysis of face detection algorithms, each method exhibited distinct strengths and weaknesses. As shown in Table 2, the Haar Cascade algorithm, while efficient with the shortest inference time among all tested methods, demonstrated moderate overall performance. Its precision and recall values indicated reasonable detection capability; however, it struggled with a higher rate of false negatives, meaning that many faces were missed. This resulted in lower recall and F1-score, suggesting that while Haar Cascade is fast, it may not be as reliable in accurately detecting all faces.

Conversely, the MTCNN model excelled in both precision and recall across various test scenarios, showcasing its robustness and accuracy in face detection. The bounding boxes produced by MTCNN were closely aligned with the ground truth annotations, reflecting high detection accuracy. Nevertheless, MTCNN's inference time was significantly longer compared to Haar Cascade, though it varied depending on the test conditions, indicating a trade-off between detection accuracy and processing speed.

YOLOFace demonstrated exemplary performance with perfect precision and recall in most scenarios. The green bounding boxes provided by YOLOFace closely matched the ground truth annotations, revealing its capability to achieve minimal false positives and false negatives. YOLOFace's superior accuracy can be attributed to its architecture, which applies a single neural network across the entire image, allowing it to simultaneously predict multiple bounding boxes and their associated probabilities. This design enables YOLOFace to detect faces more accurately and efficiently, even in complex environments, by leveraging spatial information more effectively. These findings are consistent

with the research conducted by Chen et al., who also observed YOLOFace's capability in achieving high accuracy and efficiency in face detection tasks. Chen et al. also found that YOLOFace is faster than RetinaFace, which further supports the results of this study [30].

RetinaFace also achieved perfect scores in precision and recall for most test cases, with its bounding boxes accurately detecting all faces in the images. RetinaFace's high accuracy stems from its use of a deep residual network backbone, which enhances feature extraction, and its integration of pixel-level face localization tasks. This combination allows RetinaFace to excel in detecting faces under challenging conditions, such as varying lighting or occlusions, by providing more detailed and precise feature maps [21], [22]. Despite its high accuracy, RetinaFace had the longest inference time, particularly in scenarios with diverse lighting conditions. This slower processing speed highlights a trade-off where the algorithm's high precision comes at the cost of efficiency, suggesting that RetinaFace is best suited for applications where detection accuracy is of utmost importance, even if it means accepting longer processing times.

For a more detailed analysis and clearer understanding, the following graphs provide a comprehensive visualization of each model's performance. The graphs outline key metrics such as precision, recall, F1-score, and inference time for each tested algorithm. By presenting these metrics visually, the graphs facilitate easy comparison between the different models, illustrating performance differences clearly. Each model is displayed with its corresponding metrics graphically, which helps in identifying the relative strengths and weaknesses of each algorithm in face detection.

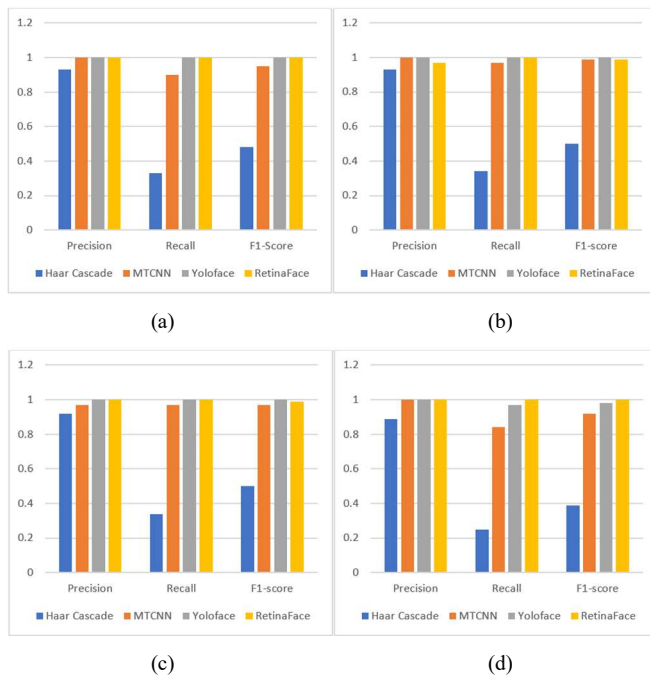


Fig 6. Evaluation metrics graph in each scenario (a) Test scenario 1, (b) Test scenario 2, (c) Test scenario 3, (d) Test scenario 4

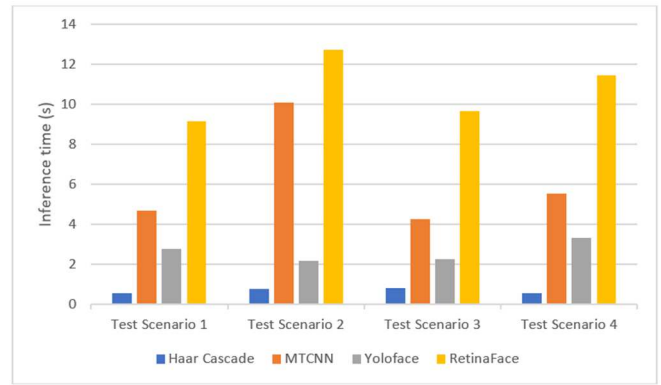


Fig 7. Inference time results for each scenario

Figures 6 and 7 summarize the performance of Haar Cascade, MTCNN, YOLOFace, and RetinaFace across different test scenarios. MTCNN and YOLOFace demonstrate consistently high precision and recall, with YOLOFace slightly leading in F1-score. RetinaFace performs well but shows longer inference times, while Haar Cascade generally exhibits lower recall and precision, particularly in challenging conditions such as low light. These figures highlight YOLOFace and MTCNN as the most reliable methods overall, with YOLOFace offering a balanced trade-off between accuracy and efficiency.

V. CONCLUSION AND FUTURE WORK

The comparative analysis of face detection algorithms in classroom environments has highlighted the distinct strengths and weaknesses of each model. YOLOFace and RetinaFace both achieved high precision and recall, with YOLOFace excelling in processing speed, making it ideal for real-time applications like automated attendance systems. RetinaFace, on the other hand, demonstrated superior accuracy in challenging conditions such as low lighting and occlusions, making it suitable for applications where precision is critical, even at the expense of speed.

MTCNN also performed well, particularly in detecting faces with varying orientations and scales, but its longer inference times may limit its use in time-sensitive scenarios. Haar Cascade, while fast, showed the lowest precision and recall, especially in complex lighting conditions, suggesting it may not be adequate for modern classroom environments where accuracy is essential.

This study underscores the need to select face detection models that are not only accurate but also adaptable to the dynamic challenges of educational settings. Future research should focus on developing face recognition models capable of identifying all students in a classroom and seamlessly integrating with other educational technologies, such as emotion recognition, engagement monitoring, and automated attendance systems.

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